

# A Causal Evaluation of Timeouts in the NBA

**Aksel Kretsinger-Walters**

*Department of Computer Science*

*Columbia University*

*New York, NY 10027, USA*

ADK2164@COLUMBIA.EDU

**Editor:** David M. Blei

## Abstract

There is a long-standing belief in the NBA that calling a timeout during an opponent’s scoring run can disrupt their momentum and improve the calling team’s chances of winning. However, the empirical evidence in support of this belief is limited and inconclusive. In this paper, we use play-by-play data from the 2021-2026 NBA seasons to conduct a causal analysis of the effect of timeouts on team performance.

**Keywords:** Causal inference, sports analytics, NBA, timeouts, momentum, difference-in-differences, synthetic control

## 1 Significance

The definition and nature of streaks, runs, and momentum itself has been a point of interest and debate for sports fans and analysts for decades. In addition to the papers cited in our related works, there’s a large body of research on the existence and nature of momentum in sports, both on the individual level (e.g. the “hot-hand” effect) and the team level.

On an academic level, the question of whether timeouts disrupt momentum is a interesting case study in causal inference in a complex, dynamic system. It involves issues of selection bias (coaches call timeouts in response to game state), confounding (many factors influence scoring dynamics), and the challenge of defining and measuring momentum itself. The conflicting conclusions in the existing literature suggest that this is a non-trivial question.

The significance of this question extends beyond academic curiosity. If timeouts do have a causal effect on disrupting momentum, it would validate a common coaching strategy and could influence how teams manage their timeouts. There’s also opportunity for further discovery: how does the relationship change conditioned on player substitutions, home-away status, quarter, regular vs post season, team age and maturity, duration and intensity of the run, etc.? There are many more such ablations to consider. Conversely, if timeouts are found to have no effect or even a negative effect, it could lead to a reevaluation of how coaches use their timeouts and potentially change the way teams approach in-game strategy.

## 2 Related Works

The question of whether timeouts actually disrupt momentum has been studied from several angles, with notably conflicting conclusions. [Assis et al. \(2020\)](#) apply a formal causal

framework to NBA play-by-play data, using matching techniques informed by a causal graph to estimate the counterfactual effect of a timeout. They conclude that timeouts have *no* effect on team performance, and argue that the apparent rebound observed after a timeout is simply regression to the mean: because timeouts tend to be called when the opposing team is scoring at an unusually high rate, any subsequent slowdown reflects the natural tendency of scoring to return to its average rather than a causal effect of the stoppage itself.

Gibbs et al. (2021) revisit the same question but restrict attention to the specific scenario of an opposing team’s scoring run, where the strategic stakes of the timeout decision are highest. Working within the Rubin causal model, they carefully define units under SUTVA and introduce an interpretable outcome based on the score difference to capture broad changes in scoring dynamics. In contrast to Assis et al. (2020), they find that calling a timeout during an opposing run is *slightly disadvantageous* on average, and further show that the magnitude of this effect varies meaningfully by franchise. Their result suggests that comebacks after runs happen frequently enough to create the illusion of timeout efficacy, but that the intervention itself does not help.

Weimer et al. (2023) take a different identification strategy by exploiting mandatory television timeouts, which are triggered at predetermined points in the game and are therefore plausibly exogenous with respect to in-game momentum. Using TV timeouts as a natural experiment that mimics random assignment of a pause, they report an *11.2% decline* in the scoring rate of the team carrying momentum immediately after a TV timeout, and they show this effect is robust to run size, substitutions, and game context. This stands in sharp contrast to the null result of Assis et al. (2020) and the mildly negative result of Gibbs et al. (2021): by removing coach selection bias entirely, Weimer et al. (2023) argue that team-level momentum genuinely exists and that pauses in play do interrupt it.

Qiu et al. (2025) approach the problem from a sports-psychology perspective rather than a causal-inference one, analyzing 4,051 timeouts across 1,235 elite basketball games using linear mixed models to test pre- versus post-timeout momentum over short (48s), medium (96s), and long (144s) windows. They find that timeouts significantly *increase* momentum for teams that are trailing or otherwise disadvantaged, while tending to *decrease* momentum for teams that are already ahead, with contextual factors such as game period and opponent strength substantially moderating the effect. Unlike the three causal papers above, which target an average treatment effect, their results emphasize heterogeneity: timeouts are neither uniformly helpful nor uniformly useless, and their efficacy is highly conditional on game state. Together these four studies motivate our own analysis, which aims to reconcile the tension between the exogenous-timeout evidence for a real momentum effect and the selection-bias-corrected evidence against one.

### 3 Methods

#### 3.1 Data

We build our analysis on the cdnnba enriched play-by-play data for the 2020–2025 NBA seasons (regular season and playoffs), loaded via our own `CDNNBAMemoPL.load_all()` pipeline. After cleaning, the spine contains 4,168,786 play-by-play rows spanning 7,457 unique games and 1,480,697 possessions, of which 82,197 are coach-logged timeouts (`actionType == timeout`, `subType` in `{full, challenge}`). Reclassifying those rows by causal trigger

(§3.2) yields 13,172 league-forced *tv\_mandatory* timeouts, 6,805 *coach\_absorb* timeouts that satisfy an imminent mandatory slot, 23,747 *coach\_discretionary* pure coach calls, 36,918 *mistagged\_discretionary* rows (qualifier-tagged as mandatory by the feed but failing the rulebook’s slot-owner / first-team-TO / proximity gates), and 1,555 coach challenges.

### 3.2 Timeout classification

We partition stoppage events into three mutually exclusive treatment groups according to whether the *coach* made a strategic choice to take the break. **Exogenous** events are stoppages that would have occurred regardless of any coach decision: *cdnnba tv\_mandatory* rows (the league-forced commercial breaks at roughly the 6:59 and 2:59 marks), *coach\_absorb* rows (a coach call within  $\sim 80$  seconds of a pending mandatory trigger, where the slot was queued to auto-fire anyway and the coach is only choosing the moment, not the fact of the timeout), and natural game pauses (`actionType == stoppage` – out-of-bounds, injuries, blood rule, equipment, etc.). **Endogenous** events are strategic coach calls – *coach\_discretionary* (no mandatory tag at all), *mistagged\_discretionary* (qualifier-tagged “mandatory” but failing the rulebook gates below), and coach challenges – restricted to those called by the team being scored against (the *suffering team*). **Control** events are dead-ball moments with no timeout occurring during an active scoring run; to avoid over-representing long runs, we subsample at most one control per run segment.

The *cdnnba* feed directly tags mandatory timeouts in its `qualifiers` column (string contains “mandatory”), so unlike prior pipelines we do not need to infer the TV-timeout slot from clock position alone. A qualifier-tagged row becomes *tv\_mandatory* only when (i) `seconds_remaining`  $\leq$  the period’s slot- $K$  trigger and (ii) the charged `teamId` matches the rulebook’s structural slot owner – home for slot 1, the opposing team for slot 2. Rows passing only the clock gate (`sr > trigger`) become *coach\_absorb* if the coach was within an 80-second buffer of the trigger *and* the row is that team’s first full TO of the period; otherwise the row is reclassified *mistagged\_discretionary* (the feed billed it as mandatory but it is structurally a coach call). The 80-second buffer is rulebook-informed but rules-based; we picked it from the empirical knee in the (`sr – trigger`) distribution and sensitivity to it is modest in the range 60–180s. To validate this procedure we ran the analogous detector on the 2013–2016 *nbastatsv3* feed – which carries explicit `Official / Official TV` ground truth – and obtained  $\mathbf{F1} = 0.933$  (precision 0.876, recall 0.999) against that label.

The endogenous/exogenous distinction is causal rather than mechanical. NBA rules allow substitutions and strategy huddles during *any* timeout (full, short, or mandatory TV), so the opportunity to alter play after the break is identical in both cases. What separates the two types is how the trigger is selected: coaches initiate endogenous timeouts in response to game state (introducing selection bias on adverse momentum), whereas mandatory TV timeouts and *coach\_absorb* events fire at fixed clock positions and are therefore plausibly exogenous conditional on the clock.

### 3.3 Run and recovery definitions

A *scoring run* at play  $t$  is defined as  $|\text{streak}(t)| \geq s$ , where  $\text{streak}(t)$  is the cumulative net score swing within the current scoring segment, computed by resetting the counter whenever

the scoring team flips. We refer to the team being outscored as the *suffering team* and the team scoring as the *running team* or alternatively the *streaking team*.

Our primary outcome is the *recovery* metric: the signed lead change from the suffering team’s perspective over a forward window of  $n$  minutes. If  $\text{lead}(t)$  is the home-minus-away margin at play  $t$ , we define

$$\text{recovery}(t, n) = -\text{sign}(\text{streak}(t)) \cdot (\text{lead}(t + n \text{ min}) - \text{lead}(t)),$$

so that positive values mean the suffering team clawed points back during the window. As a secondary outcome we also report `running_team_pts`, the raw points scored by the running team over the forward window, which matches the dependent variable used by [Weimer et al. \(2023\)](#).

### 3.4 Between-group estimators

As a first pass we compute group means of `recovery` conditional on run size  $s$  and forward window  $n$  (measured in minutes), and report differences versus the control group using Welch’s  $t$ -tests. We sweep  $s \in \{4, 5, 6, 7, 8, 10, 12, 15\}$  and  $n \in \{0.5, 1, 1.5, 2, 3, 4, 5\}$  minutes, and also stratify by quarter, regular season vs. playoffs, and absolute score margin.

### 3.5 Within-game matched-twin causal design

**Motivation.** The between-group estimator is vulnerable to game-level confounding: coaches call timeouts in systematically different games than the ones where control events are drawn, and mandatory TV timeouts cluster at predictable game-clock positions (around the 6:59 and 2:59 marks of each period) rather than uniformly across all game contexts. Comparing the timeout pool against a global control mean therefore absorbs game- and clock-level selection into the baseline. To remove this bias we use a *within-game matched-twin* design that holds team composition, pace, coaching, and game-level momentum approximately fixed by construction.

**Method.** We first build a candidate pool by classifying every spine event during a run of  $|\text{streak}| \geq s$  into one of three groups: **treated-endogenous** (a coach `full` or `challenge` timeout called by the suffering team), **treated-exogenous** (an `official_inferred` TV timeout or a generic dead-ball `stoppage`), or **control** (any spine event during the same run that is not classified as treated, with at most one control sampled per contiguous run segment to avoid over-counting long runs). For each treated event we then search the control pool *within the same game* for a candidate that satisfies all of: the same period; the same sign of `streak` (so the run is against the same team);  $|\Delta \text{streak}| \leq 2$  (comparable run magnitude);  $|\Delta \text{seconds\_remaining}| \leq 120$  s (comparable clock position within the period); and  $|\Delta \text{suffering\_margin}| \leq 3$ , where the suffering margin

$$\text{suffering\_margin}(t) = -\text{sign}(\text{streak}(t)) \cdot \text{lead}(t)$$

is the score margin from the suffering team’s perspective (positive when the suffering team is still ahead despite the run against it). When multiple candidate controls satisfy the constraints, we select the one minimising the weighted Manhattan distance

$$d(\text{treated}, \text{control}) = 2|\Delta \text{streak}| + \frac{|\Delta \text{seconds\_remaining}|}{30} + 3|\Delta \text{suffering\_margin}|,$$

which trades a  $\sim 30$ -second clock gap for one point of streak magnitude or one point of score margin. Each treated event yields at most one pair; the same control row may serve as the matched twin for multiple treated events. The per-pair causal estimate is the difference in **recovery** between the treated event and its matched twin, and we aggregate across pairs with a paired  $t$ -test.

### 3.6 Moderator analyses

To probe heterogeneity we repeat the between-group and matched-twin analyses within slices defined by (i) timeout subtype (**tv\_mandatory**, **coach\_absorb**, **coach\_discretionary**, **mistagged\_discretionary**, **coach\_challenge**, **stoppage**); (ii) game phase and clutch status (last five minutes of Q4 with margin  $\leq 5$ ); (iii) season-long team quality gap, using each team’s **NET\_RATING** from **player\_advanced\_stats**; (iv) the number of substitutions within a  $\pm 30$ s game-clock window around the event, bucketed as  $\{0, 1-2, 3+\}$ ; and (v) the running team’s in-game cumulative points-per-possession (PPP) at the time of the event, as a proxy for short-term shooting rhythm.

## 4 Evaluation

### 4.1 Between-group results

The simple between-group comparison essentially reproduces the weak-to-null effects reported in the prior literature. Across run sizes  $s \in \{5, \dots, 10\}$  and forward windows  $n \in \{1, \dots, 5\}$  minutes, the endogenous-vs-control gap in **recovery** is never statistically significant (all  $|\Delta| < 0.1$  pts,  $p > 0.05$  in 15 of 16 cells). Exogenous timeouts, by contrast, look significantly *negative*, with  $\Delta \approx -0.25$  to  $-0.45$  pts at the 3-minute window depending on run size ( $p < 0.001$ ). Sweeping the forward window at fixed run size  $s = 6$  shows the exogenous penalty growing from  $-0.20$  at 30 seconds to a plateau around  $-0.27$  at 3 minutes, while the endogenous gap hovers near zero throughout. At face value this matches the “timeouts don’t help (and mandatory TV breaks hurt)” consensus. As a secondary check we also computed **running\_team\_pts** in the style of Weimer et al. (2023) and found a *positive* exogenous-minus-control gap of  $+2.8\%$  to  $+14.6\%$  depending on window – the opposite sign from Weimer’s reported  $-11.2\%$ . This is internally consistent with our suffering-team analysis (if the suffering team recovers less, the running team must score relatively more) and likely reflects the post-2017 mandatory-timeout rule changes together with differences in how we classify mandatory timeouts.

To avoid confusion we restate the result: under a between-group estimator, endogenous coach timeouts look indistinguishable from control and exogenous breaks look mildly negative. We believe this is **not** a causal effect but instead a consequence of the adverse situations in which coaches predominantly call timeouts – a game-level selection bias that is absorbed into the global control baseline. The matched-twin design of §4.2, which holds game identity (and therefore that selection) fixed by construction, will reverse this conclusion and produce a strongly positive and *symmetric* causal estimate for both endogenous and exogenous breaks.

## 4.2 Within-game matched-twin results

The matched-twin design of §3.5 – our cleanest causal test – reverses the headline conclusion of the between-group analysis. With  $s = 6$  and a 3-minute window we obtain 7,509 matched endogenous pairs and 3,480 matched exogenous pairs (each treated event matched to its closest in-game control by the weighted distance  $d$ ; control rows are reused across treated rows when they are the best match for multiple). The paired mean differences are **+0.410 pts for endogenous timeouts** ( $t = 26.6$ ,  $p < 0.0001$ ) and **+0.405 pts for exogenous timeouts** ( $t = 13.2$ ,  $p < 0.0001$ ). Both effects are highly significant, both favor the suffering team, and – notably – they are *statistically indistinguishable in magnitude*. Breaking the treated population into the six fine subtypes of §3.2 reveals the same sign and roughly the same magnitude everywhere: **tv\_mandatory** +0.361 \*\*\* ( $n = 879$ ), **generic stoppage** +0.458 \*\*\* ( $n = 1,752$ ), **coach\_absorb** +0.340 \*\*\* ( $n = 849$ ), **coach\_discretionary** +0.428 \*\*\* ( $n = 2,119$ ), **mistagged\_discretionary** +0.400 \*\*\* ( $n = 5,355$ ), and **coach\_challenge** +0.714 \* ( $n = 35$ ). The consistency across six structurally distinct event types – league-forced auto-fires, natural stoppages, last-second coach absorbs, pure discretionary calls, mistagged-as-mandatory coach calls, and replay challenges – is itself evidence that the effect is driven by the pause itself, not by who chose to take it or how the league billed it.

The reversal relative to the between-group results is driven by game-level selection: coach timeouts concentrate in games where the suffering team historically claws back less of its deficit during opponent runs (low-recovery games), so comparing the endogenous mean to a global control mean absorbs that selection into the baseline and erases the signal. The matched twin lives in the same game, same period, with the same streak sign and a near-identical clock position and score margin, so any game-level confounder (team quality, pace, coaching style, overall momentum direction) is differenced out at the pair level and the estimator survives whatever selection-on-game produced the negative exogenous gap in §4.1. A coarser within-game *mean* comparison – in which each treated event is compared against the average **recovery** of all controls in the same game, rather than against a single distance-minimising twin – gives a consistent +0.35 pts effect for endogenous timeouts but shrinks the exogenous effect to near zero. The drop suggests that tight clock-position matching is more load-bearing for exogenous events (whose triggers are themselves clock-locked) than for endogenous ones.

The +0.714 pair difference on coach challenges is large in magnitude but rests on  $n = 35$  matched pairs and reaches only  $p < 0.05$ . Two structural factors limit how much weight we can put on this row. First, the cdnnba feed only emits **instantreplay** outcome rows for the 2020 and 2021 seasons; challenges from 2022 onward have no recorded **overturned** / **stands** label, so we cannot examine the broader population. Second, even when we restrict to those two seasons and try to split the matched challenges by replay outcome (E15), only 19 of the 35 challenges line up with a replay row within our 60-second window and – coincidentally or otherwise – all 19 are **overturned**, leaving zero failed challenges in the matched sample. The broader 2020–2021 challenge population (1,446 challenges from **instantreplay/challenge** rows, of which 688 are overturned and 758 fail) shows a near-even success/failure split, so the all-overturned matched sample is a sampling artifact rather than a true asymmetry. We therefore cannot test the natural mechanistic hypothesis – that successful challenges



drive the entire coach\_challenge effect – with the present matching protocol; we flag it as future work (§4.4) for a redesigned challenge-specific match that loosens the per-period clock constraint.

### 4.3 Mechanism and moderators

Once we have a signal, we can ask where it comes from. All moderator analyses in this section use a between-group Welch test of treated vs control within each stratum – the same estimator as §4.1, not the paired matched-twin of §4.2. The magnitudes below are therefore directly comparable to the near-zero between-group baseline and reflect the residual signal that survives the selection bias after stratifying, not the underlying causal effect size; we use the coarser estimator here to keep per-stratum sample sizes large enough for a clean test.

Stratifying by the number of substitutions in a  $\pm 30$ s window around the event reveals that the endogenous effect is concentrated entirely in timeouts accompanied by personnel changes: with 0 substitutions the endogenous-control gap is  $-0.05$  (n.s.), with 1–2 substitutions it is  $+0.10$  (n.s.), and with 3 or more substitutions it rises to  $+0.21$  ( $p < 0.001$ ). The exogenous penalty is correspondingly largest ( $-0.39$ ,  $p < 0.001$ ) when no subs occur and vanishes when many do. This directly confirms the mechanism proposed by [Weimer et al. \(2023\)](#): the value of a timeout is mediated by substitutions rather than by the pause itself. Conditioning on team quality via season-long `NET_RATING`, we find that the endogenous benefit is strongest in evenly matched games ( $+0.23$  pts,  $p < 0.05$ ) and washes out in lopsided matchups where the talent gap dominates the trajectory. Finally, stratifying by the running team’s in-game PPP we find that when the opponent is shooting unusually efficiently ( $PPP \geq 1.30$ ), both effects are amplified and point in opposite directions: endogenous timeouts produce  $+0.24$  pts of extra recovery ( $p < 0.05$ ) while exogenous timeouts produce  $-0.62$  pts ( $p < 0.001$ ). This is the clearest evidence in our data of a genuine momentum effect – a coach timeout with substitutions disrupts a hot-shooting opponent, while a pure TV break effectively gives that opponent a rest and lets the run continue.

### 4.4 Summary

Our headline finding is that timeouts causally improve the suffering team’s short-term recovery by roughly  $+0.4$  points over the following three minutes, and that this effect is masked in between-group analyses by a subtle but severe game-level selection bias. The within-game matched-twin estimator, which holds game identity approximately fixed by construction, produces a strongly positive and highly significant effect of  $+0.410$  **pts** for endogenous (strategic coach) breaks and  $+0.405$  **pts** for exogenous (league-forced or auto-fire) breaks – two estimates that are statistically indistinguishable. That symmetry, together with the consistent  $+0.34$  to  $+0.46$  pair-difference across six structurally distinct fine subtypes, is strong evidence that the effect is driven by the pause itself rather than by who chose to take it. The published consensus of “no effect” or “slightly negative effect” ([Assis et al., 2020](#); [Gibbs et al., 2021](#)) reproduces in our data only under a between-group estimator that absorbs the game-level selection into the baseline. The benefit is overwhelmingly mediated by substitutions (in agreement with [Weimer et al. \(2023\)](#)) and is largest in evenly matched

games and against hot-shooting opponents, which matches both coaching intuition and the conditional-momentum findings of [Qiu et al. \(2025\)](#).

## Future Work

Several extensions follow naturally from the present design:

- **Better pairing mechanism.** How should we score the quality of a matched twin? The current Manhattan-weighted distance  $d$  is a reasonable first cut, but there is a sweet-spot to characterise between pair similarity (which tightens the estimate) and matched-pair sample size (which determines its precision).
- **Alternative outcome metrics.** Beyond **recovery**, we could measure the streaking team’s points-per-possession or points-per-minute over the forward window, or roll either up into a win-probability impact via a calibrated WP model.
- **Causal graphical model.** Distil the moderator findings into an explicit CGM (substitutions  $\rightarrow$  recovery; running-team PPP  $\rightarrow$  recovery; **NET\_RATING** gap  $\rightarrow$  recovery) and use it to formalise the identification arguments and check for missing confounders.
- **Streaking-team timeouts.** Re-run the analysis with the *streaking* team as the calling team. Do they substitute players at a higher rate than baseline, and what mechanism would explain a positive or negative effect on the run they are already riding?
- **Additional break types.** Official reviews, injury stoppages, and ejections all introduce small exogenous perturbations to play. Do they show the same recovery pattern as TV timeouts? Cross-checking against more natural experiments would strengthen the causal claim.

## Further experimentation and code changes

A running log of follow-up items raised during the analysis. Items marked **[done]** have been folded into the current draft; the rest remain open.

- **[done]** Coach challenge split by successful / failed outcome (E15). Implemented in `run_experiments.v2.experiment_15` and invoked from the notebook; results discussed in §4.2. Matched-sample size collapses to 7 challenge pairs (19 of which are overturned, 0 failed), so we cannot test the success/failure asymmetry with the present per-period match constraint.
- **[done]** Reconcile coach\_challenge counts between the matched-twin section and E15. The 35 challenges in the matched sample drop to 19 with replay outcomes attached because the cdnba feed only emits `instantreplay` rows for the 2020–2021 seasons; from 2022 onward replay outcomes are absent. Pre-2020 and post-2021 challenges are present as timeout rows but cannot be split by outcome.



- **[done]** Mandatory-timeout classifier. The previous inferred-injection pipeline was replaced with a cause-based reclassifier driven by the `cdnnba qualifiers` field, with strict rulebook gates on slot owner (home / away) and clock proximity. F1 against the v3 ground-truth label rose from 0.88–0.92 to 0.933.
- **[done]** Reconcile values between sections 4.2 and 4.3 after the new mandatory classification. Numbers in both sections are now produced from the same rerun.
- **[done]** Ensure only timeouts called by the suffering team enter the endogenous filter. The `suffering_called` mask was already present in `build_rich_analysis`; verified after the cause-taxonomy refactor.
- **[done]** Re-run the experiment battery after the classifier and cause-taxonomy changes, flag differences, update conclusions. The post-rerun headline shifted from “endogenous  $\approx 0$ , exogenous  $\approx -0.27$ ” (between-group) to “endogenous and exogenous both  $\approx +0.4$ ” (matched-twin) – this is now the report’s headline.
- *(open)* Use real-world elapsed time (`timeActual`) as a feature in the TV-timeout injection model and retrain/evaluate on nbastatsv3. The current `cdnnba` pipeline does not need it (the `qualifiers` field carries the structural signal directly), but it might strengthen the pre-2017 v3 detector.
- *(open)* Analyse real-world elapsed time as a mechanism: do longer breaks correlate with larger recovery? `timeout_duration_s` is persisted on the dataset and this is a one-cell experiment.
- *(open)* Team age as a moderator. Use the average age of players on the court at the time of the break, plus the delta in average age before vs after the break (a proxy for whether the coach swapped out tired veterans for fresh legs).
- *(open)* End-of-quarter trimming: never include events in the last  $k$  (say,  $k = 2$ ) minutes of a period, and only include forward-window outcomes that fit inside the period (e.g. with 3.5 minutes left, accept the 3-minute window but exclude the 4-minute one). Would tighten the controls but cost sample size.
- *(open)* Update NBA data through the current season. Expensive: Ted’s data is not easily redistributable, so this requires re-scraping the `cdnnba` feed end-to-end.
- *(open)* Investigate simple substitutions without official timeouts (e.g. at free throws). When else are teams allowed to substitute? A clean substitution-without-pause control would isolate the personnel-change mechanism from the pause itself.
- *(open)* Re-run with the forward window measured in *possessions* instead of minutes, ensuring the window contains an even number of possessions so each team gets the same opportunity count (modulo the rare technical-foul edge case).
- *(open)* Reconcile with the Weimer finding of a  $-11.2\%$  drop in streaking-team scoring during exogenous breaks. Our suffering-team estimate already inverts the sign in the between-group view; a head-to-head replication that mirrors Weimer’s exact definition is deferred.

## Acknowledgments and Disclosure of Funding

N/A

## LLM Usage

`claudeopus-4.6` was utilized extensively during the implementation of this research project, and we wanted to acknowledge its contributions.

The author(s) were responsible for project idea generation, sourcing relevant papers and publications, research roadmap, code scaffolding, and report skeleton.

`Opus` assisted in the implementation: data preprocessing and cleaning, feature generation, significance testing, ablation studies, and report completion. All `Opus` work was closely reviewed and cross-examined.

## References

- Niander Assis, Renato Assunção, and Pedro O. S. Vaz-de Melo. Stop the clock: Are timeout effects real? In *Machine Learning and Knowledge Discovery in Databases (ECML PKDD)*, 2020. URL <https://arxiv.org/abs/2009.06750>. arXiv:2009.06750.
- Connor Gibbs, Ryan Elmore, and Bailey Fosdick. The causal effect of a timeout at stopping an opposing run in the NBA. *arXiv preprint arXiv:2011.11691*, 2021. URL <https://arxiv.org/abs/2011.11691>.
- Mingjia Qiu, Kaimin Zhang, Yafei Chao, and Mingxin Zhang. Interrupt or reinforce? the impact of timeout on momentum in basketball game. *Frontiers in Psychology*, 16:1673186, 2025. doi: 10.3389/fpsyg.2025.1673186. URL <https://doi.org/10.3389/fpsyg.2025.1673186>.
- Louis Weimer, Zachary C. Steinert-Threlkeld, and Kevin Coltin. A causal approach for detecting team-level momentum in NBA games. *Journal of Sports Analytics*, 9(2):117–132, 2023. doi: 10.3233/JSA-220592. URL <https://doi.org/10.3233/JSA-220592>.

## Appendix

Tables below reproduce the post-rerun E9–E14 results from the analysis pipeline driver (`run_experiments.v2.py`) on the `cdnba` dataset, run size  $s = 6$ , 3-minute forward window, with the cause-based reclassifier and strict slot-owner gate of §3.2. Significance codes: n.s. =  $p \geq 0.05$ , \* =  $p < 0.05$ , \*\* =  $p < 0.01$ , \*\*\* =  $p < 0.001$ .

### E9: Matched-twin causal estimates

| group      | pairs | treated $\mu$ | matched ctrl $\mu$ | pair diff | sig |
|------------|-------|---------------|--------------------|-----------|-----|
| endogenous | 7,509 | +0.352        | −0.058             | +0.410    | *** |
| exogenous  | 3,480 | +0.198        | −0.207             | +0.405    | *** |

  

| subtype                 | pairs | treated $\mu$ | matched ctrl $\mu$ | pair diff | sig |
|-------------------------|-------|---------------|--------------------|-----------|-----|
| tv_mandatory            | 879   | +0.259        | −0.101             | +0.361    | *** |
| stoppage                | 1,752 | +0.059        | −0.399             | +0.458    | *** |
| coach_absorb            | 849   | +0.422        | +0.081             | +0.340    | *** |
| coach_discretionary     | 2,119 | +0.352        | −0.076             | +0.428    | *** |
| mistagged_discretionary | 5,355 | +0.349        | −0.051             | +0.400    | *** |
| coach_challenge         | 35    | +0.800        | +0.086             | +0.714    | *   |

**E10: Between-group recovery by fine subtype**

| subtype                 | $n$    | $\mu$  | ctrl $\mu$ | $\Delta$ vs ctrl (sig) |
|-------------------------|--------|--------|------------|------------------------|
| tv_mandatory            | 1,486  | +0.283 | +0.371     | -0.087 n.s.            |
| stoppage                | 3,793  | +0.125 | +0.371     | -0.246 ***             |
| coach_absorb            | 1,563  | +0.460 | +0.371     | +0.089 n.s.            |
| coach_discretionary     | 4,694  | +0.432 | +0.371     | +0.062 n.s.            |
| mistagged_discretionary | 11,543 | +0.430 | +0.371     | +0.059 n.s.            |
| coach_challenge         | 70     | -0.429 | +0.371     | -0.799 n.s.            |

**E11: Game phase and clutch conditioning**

| condition                            | endo $n$ | endo $\mu$ | exo $\mu$ | ctrl $\mu$ | $\Delta$ endo-ctrl | $\Delta$ exo-ctrl |
|--------------------------------------|----------|------------|-----------|------------|--------------------|-------------------|
| Q1 early (0–360s)                    | 1,415    | +0.408     | +0.327    | +0.470     | -0.062 n.s.        | -0.143 n.s.       |
| Q1 late (360–720s)                   | 1,242    | +0.554     | +0.322    | +0.386     | +0.168 n.s.        | -0.063 n.s.       |
| Q2 early (720–1080s)                 | 2,451    | +0.585     | +0.303    | +0.470     | +0.114 n.s.        | -0.167 n.s.       |
| Q2 late (1080–1440s)                 | 1,524    | +0.295     | +0.148    | +0.283     | +0.011 n.s.        | -0.136 n.s.       |
| Q3 early (1440–1800s)                | 2,517    | +0.402     | +0.289    | +0.308     | +0.094 n.s.        | -0.018 n.s.       |
| Q3 late (1800–2160s)                 | 1,694    | +0.315     | +0.144    | +0.407     | -0.091 n.s.        | -0.262 n.s.       |
| Q4 early (2160–2520s)                | 2,710    | +0.435     | +0.089    | +0.317     | +0.119 n.s.        | -0.228 n.s.       |
| Q4 late (2520–2880s)                 | 2,584    | +0.390     | +0.100    | +0.272     | +0.118 n.s.        | -0.172 n.s.       |
| Clutch (Q4 last 5min, $ m  \leq 5$ ) | 1,006    | +0.257     | -0.253    | +0.158     | +0.100 n.s.        | -0.411 n.s.       |
| Non-clutch                           | 15,301   | +0.438     | +0.249    | +0.378     | +0.060 n.s.        | -0.129 *          |

**E12: Team quality conditioning (NET\_RATING)**

| condition                                | endo $n$ | endo $\mu$ | exo $\mu$ | ctrl $\mu$ | $\Delta$ endo-ctrl | $\Delta$ exo-ctrl |
|------------------------------------------|----------|------------|-----------|------------|--------------------|-------------------|
| Sufferer much worse ( $\Delta \leq -5$ ) | 4,113    | -0.153     | -0.134    | -0.224     | +0.071 n.s.        | +0.090 n.s.       |
| Sufferer worse ( $-5 < \Delta \leq -1$ ) | 3,830    | +0.312     | +0.013    | +0.221     | +0.091 n.s.        | -0.208 n.s.       |
| Evenly matched ( $ \Delta  < 1$ )        | 2,409    | +0.651     | +0.170    | +0.341     | +0.310 **          | -0.170 n.s.       |
| Sufferer better ( $1 \leq \Delta < 5$ )  | 3,345    | +0.589     | +0.453    | +0.556     | +0.033 n.s.        | -0.103 n.s.       |
| Sufferer much better ( $\Delta \geq 5$ ) | 2,610    | +1.096     | +0.924    | +0.986     | +0.110 n.s.        | -0.062 n.s.       |

**E13: Substitution-adjusted analysis**

| $\pm 30$ s subs | endo $n$ | endo $\mu$ | exo $\mu$ | ctrl $\mu$ | $\Delta$ endo-ctrl | $\Delta$ exo-ctrl |
|-----------------|----------|------------|-----------|------------|--------------------|-------------------|
| 0 subs          | 5,202    | +0.424     | +0.274    | +0.489     | -0.065 n.s.        | -0.215 *          |
| 1–2 subs        | 3,886    | +0.342     | +0.160    | +0.237     | +0.105 n.s.        | -0.077 n.s.       |
| 3+ subs         | 7,219    | +0.475     | +0.248    | +0.262     | +0.213 ***         | -0.014 n.s.       |

**E14: Running-team in-game PPP**

| condition               | endo $n$ | endo $\mu$ | exo $\mu$ | ctrl $\mu$ | $\Delta$ endo-ctrl | $\Delta$ exo-ctrl |
|-------------------------|----------|------------|-----------|------------|--------------------|-------------------|
| Running PPP $< 1.00$    | 4,754    | +0.345     | +0.284    | +0.145     | +0.200 *           | +0.139 n.s.       |
| Running PPP 1.00–1.15   | 5,775    | +0.420     | +0.238    | +0.341     | +0.079 n.s.        | −0.103 n.s.       |
| Running PPP 1.15–1.30   | 4,171    | +0.443     | +0.255    | +0.420     | +0.023 n.s.        | −0.164 n.s.       |
| Running PPP $\geq 1.30$ | 1,607    | +0.656     | +0.104    | +0.474     | +0.182 n.s.        | −0.370 *          |